Capstone failure testing

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Grounded Learning Service

THE CONNECTED WEEK

One complete failure matrix and regression record.

Keep expected states, claims, allowed/forbidden calls, observed traces, coverage and mandatory gates together.

Keep one habit: Recall → W1–W3/W5 → independent six-case gate → failure-testing project → delayed return.

MAIN LESSON · SECTION 1

Recall · Retrieve before notes

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Capstone service implementation. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Red team exercises. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Accessibility and user feedback. Give an example and a mistake that the idea helps you avoid.

- Use this reference for the original W1–W3 and W5 exercises. The separately named independent case supplies its own changed facts; use those facts for that attempt before feedback.

- Capstone failure testing checks denied roles, missing **evidence**, invalid **input**, timeouts and stale data. Each expected outcome is written before the run.

- Use a **failure matrix** linking condition to expected response and forbidden calls. Test both visible result and operation trace. Preserve failed fixtures for regression checks.

Capstone service implementation

> **My remembered idea, example and correction**
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Red team exercises

> **My remembered idea, example and correction**
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Accessibility and user feedback

> **My remembered idea, example and correction**
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Lesson reference for workbook practice

MAIN LESSON · SECTION 5

W1 · Explain the current example

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Failure testing declares expected outputs and forbidden operations before the run. Complete main matrix: malformed A with extra role field → invalid_input/null claims/no lookup; student T → denied/null/no lookup; absent Z → review/null/lookup Z; approved current A c1 → supported 12/REG-A/lookup A. All four pass when response, claims and exact trace agree. Denial/review can be correct outcomes.

  Deliberate bad-order T trace contains lookup T before denial, so that case fails and only 3/4 pass in the separate counterexample. Preserve original failed trace and later regression evidence. Separate picture: current approved F4 says club time 15:00, drafting unavailable, checked template explicitly allowed → supported “The club time is 15:00.”/F4 with degraded mode.

  If retrieval also fails, or evidence invalid/unapproved, review with null claim/source; template cannot invent evidence. Code uses local flags/dictionaries, no actual restricted reads, model calls or provider failures. Four main cases plus added probes are not every timeout/stale/interaction test. Mandatory gates remain independent of aggregate pass rates.

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

My reasoning and evidence

MAIN LESSON · SECTION 6

W2 · Trace the calculation

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Failure testing declares expected outputs and forbidden operations before the run. Complete main matrix: malformed A with extra role field → invalid_input/null claims/no lookup; student T → denied/null/no lookup; absent Z → review/null/lookup Z; approved current A c1 → supported 12/REG-A/lookup A. All four pass when response, claims and exact trace agree. Denial/review can be correct outcomes.

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  If retrieval also fails, or evidence invalid/unapproved, review with null claim/source; template cannot invent evidence. Code uses local flags/dictionaries, no actual restricted reads, model calls or provider failures. Four main cases plus added probes are not every timeout/stale/interaction test. Mandatory gates remain independent of aggregate pass rates.

- Four cases: malformed request, student teacher-guide request, missing register, valid current record. One rejects invalid input, one denies access, one reviews and one supports; no restricted reads occur. Show the intermediate steps and give the conclusion with its scope.

> **My reasoning and evidence**

MAIN LESSON · SECTION 7

W3 · Repair a misconception

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Failure testing declares expected outputs and forbidden operations before the run. Complete main matrix: malformed A with extra role field → invalid_input/null claims/no lookup; student T → denied/null/no lookup; absent Z → review/null/lookup Z; approved current A c1 → supported 12/REG-A/lookup A. All four pass when response, claims and exact trace agree. Denial/review can be correct outcomes.

  Deliberate bad-order T trace contains lookup T before denial, so that case fails and only 3/4 pass in the separate counterexample. Preserve original failed trace and later regression evidence. Separate picture: current approved F4 says club time 15:00, drafting unavailable, checked template explicitly allowed → supported "The club time is 15:00."/F4 with degraded mode.

  If retrieval also fails, or evidence invalid/unapproved, review with null claim/source; template cannot invent evidence. Code uses local flags/dictionaries, no actual restricted reads, model calls or provider failures. Four main cases plus added probes are not every timeout/stale/interaction test. Mandatory gates remain independent of aggregate pass rates.

- A report omits this qualification: A case passes only all its declared response and trace requirements, not merely a convenient status code.

  Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

> **My reasoning and evidence**

MAIN LESSON · SECTION 9

W5 · Explain the limitation

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Failure testing declares expected outputs and forbidden operations before the run. Complete main matrix: malformed A with extra role field → invalid_input/null claims/no lookup; student T → denied/null/no lookup; absent Z → review/null/lookup Z; approved current A c1 → supported 12/REG-A/lookup A. All four pass when response, claims and exact trace agree. Denial/review can be correct outcomes.

  Deliberate bad-order T trace contains lookup T before denial, so that case fails and only 3/4 pass in the separate counterexample. Preserve original failed trace and later regression evidence. Separate picture: current approved F4 says club time 15:00, drafting unavailable, checked template explicitly allowed → supported "The club time is 15:00."/F4 with degraded mode.

  If retrieval also fails, or evidence invalid/unapproved, review with null claim/source; template cannot invent evidence. Code uses local flags/dictionaries, no actual restricted reads, model calls or provider failures. Four main cases plus added probes are not every timeout/stale/interaction test. Mandatory gates remain independent of aggregate pass rates.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

Try alone · Selected transfer

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Six cases: five expected outcomes pass; one denied request still performs restricted lookup. Compute case pass rate and release implication.

- Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Independent case: Apply a new case independently

> **My independent explanation, trace and limits**

MAIN LESSON · SECTION 10

Project · Keep the service evidence

> Preserve the contract, inputs, observed outcomes and limits.

Complete task and evidence

- For Grounded Learning Service, use this week's independent task as a rehearsal for the final **evidence** pack: Six cases: five expected outcomes pass; one denied request still performs restricted lookup.

  Compute case pass rate and release implication. Keep actual **observation**s separate from expected results, preserve failures, compare a baseline and explain the limits.

> **My project change, supporting evidence and remaining limit**

MAIN LESSON · SECTION 11

Feedback · Repair and return

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	

Evidence	My record
What I can now explain without notes	
One remaining question	

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use W2: Trace the worked case for Capstone failure testing. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

▼ Extra practice · Individual reasoning check

REFERENCE · SECTION 3

Extra practice · Individual reasoning check

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use the worked case for Capstone failure testing. Proposed explanation: "The worked result establishes every future case". Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Earlier W4 response copy

REFERENCE · SECTION 8

Reference · Earlier W4 response copy

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Six cases: five expected outcomes pass; one denied request still performs restricted lookup. Compute case pass rate and release implication.

> **My reasoning and evidence**

▼ **Extra practice · Pictured case**

REFERENCE · SECTION 12

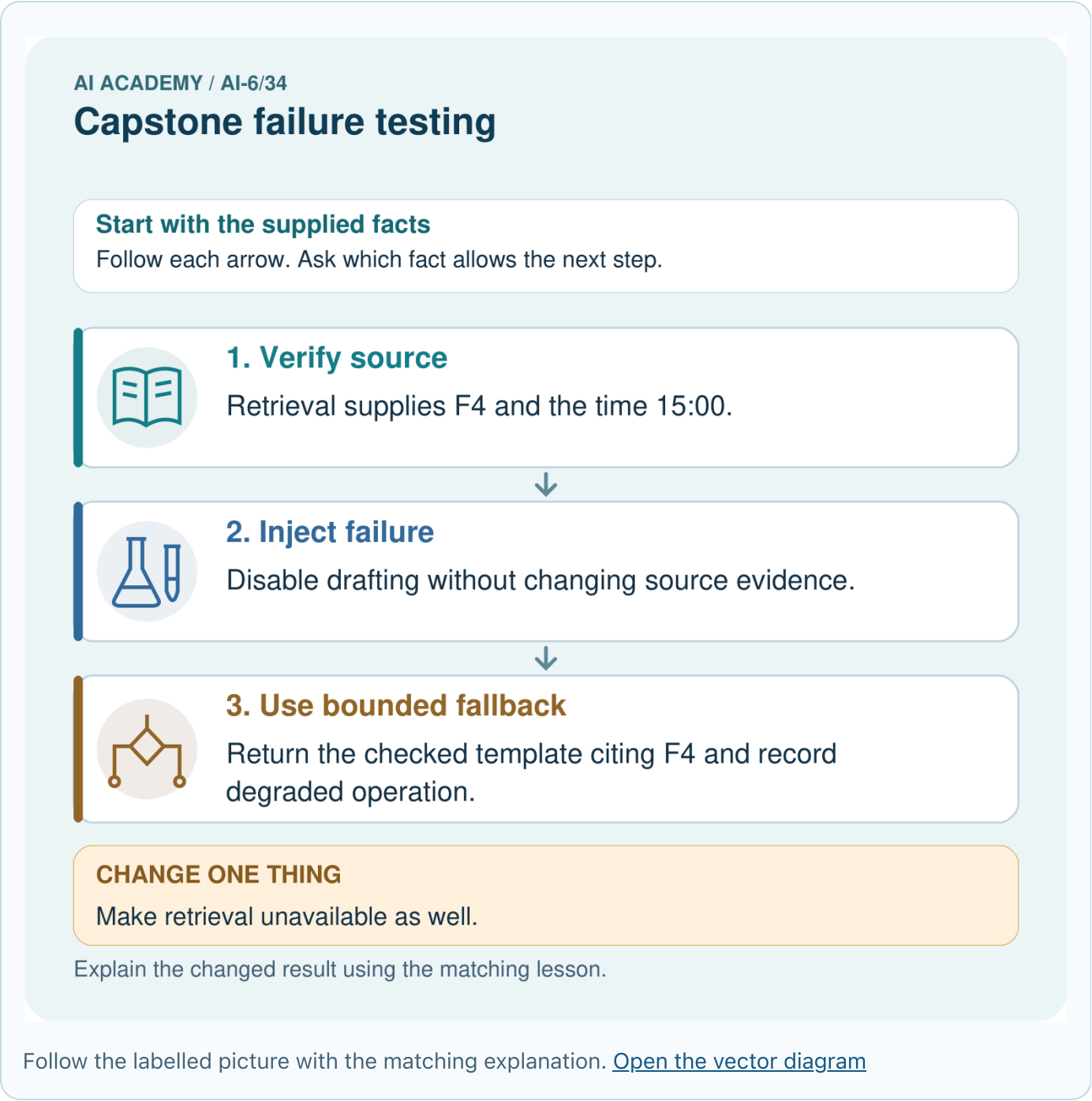
Extra practice · Pictured case

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Inject one component failure. A capstone test blocks model drafting after a successful approved-card retrieval.
- F4 records club time 15:00. The contract permits a checked fixed-template answer when drafting is unavailable.
- Explain why retrieval success permits the checked 15:00 fallback when drafting fails. What changes if retrieval fails too, and why can a template not fill that **evidence** gap?
- Support used: alone / hint / worked example. This record is practice **evidence**.

- Project connection: For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.



Follow the labelled picture with the matching explanation. Open the vector diagram

My picture-based explanation and changed-case prediction

▼ **Reference · Project milestone**

REFERENCE · SECTION 13

Reference · Project milestone

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Project: Grounded Learning Service
- Current milestone: Present and defend. Demonstrate the system, explain one failure and justify the next improvement with **evidence**.
- For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	

Evidence to keep	My record
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ **Extra practice · Predict,check and repair**

REFERENCE · SECTION 14

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- Inject one component failure. A capstone test blocks model drafting after a successful approved-card retrieval.
- Which labelled step matches this detail? Return the checked template citing F4 and record degraded operation.
- 1. Verify source
- 2. Use bounded fallback

- 3. Inject failure

- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: Use bounded fallback Return the checked template citing F4 and record degraded operation.

Failure injection changes a component deliberately so the team can inspect the response. A fallback must retain **evidence** checks and honest status.

Changed-case question and optional hint

- Changed-case practice: Make retrieval unavailable as well. Which conclusion is supported by the supplied facts?

- 1. We can decide the result without examining the changed **input** or the procedure.

- 2. Return unsupported or unavailable; do not use the factual template.

- 3. This one example guarantees correct results for every future **input**.

- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: Return unsupported or unavailable; do not use the factual template. A template can phrase **evidence** but cannot replace missing **evidence**.

Repair and retain the record

Review Capstone service implementation (ai-6/33). Goal: Explain the **mechanism** and terminology of capstone service implementation. Short practice: Trace the G2 fact request and edit request through the prototype.

What can be returned, what cannot be done, and how does removing G2 change those two outcomes?. Return to this week and explain what you changed.

Review Red team exercises (ai-6/31). Goal: Explain the **mechanism** and terminology of red team exercises. Short practice: Distinguish the exhibit fact from the embedded edit instruction.

Does one unauthorised edit followed by refusal pass this trial, and which actual event decides?. Return to this week and explain what you changed.

Review Accessibility and user feedback (ai-6/28). Goal: Explain the **mechanism** and terminology of accessibility and user feedback.

Short practice: Explain what each text, keyboard and assistive-technology change adds to the feedback. Why does changing the red circle to orange alone not solve the communication barrier?.

Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

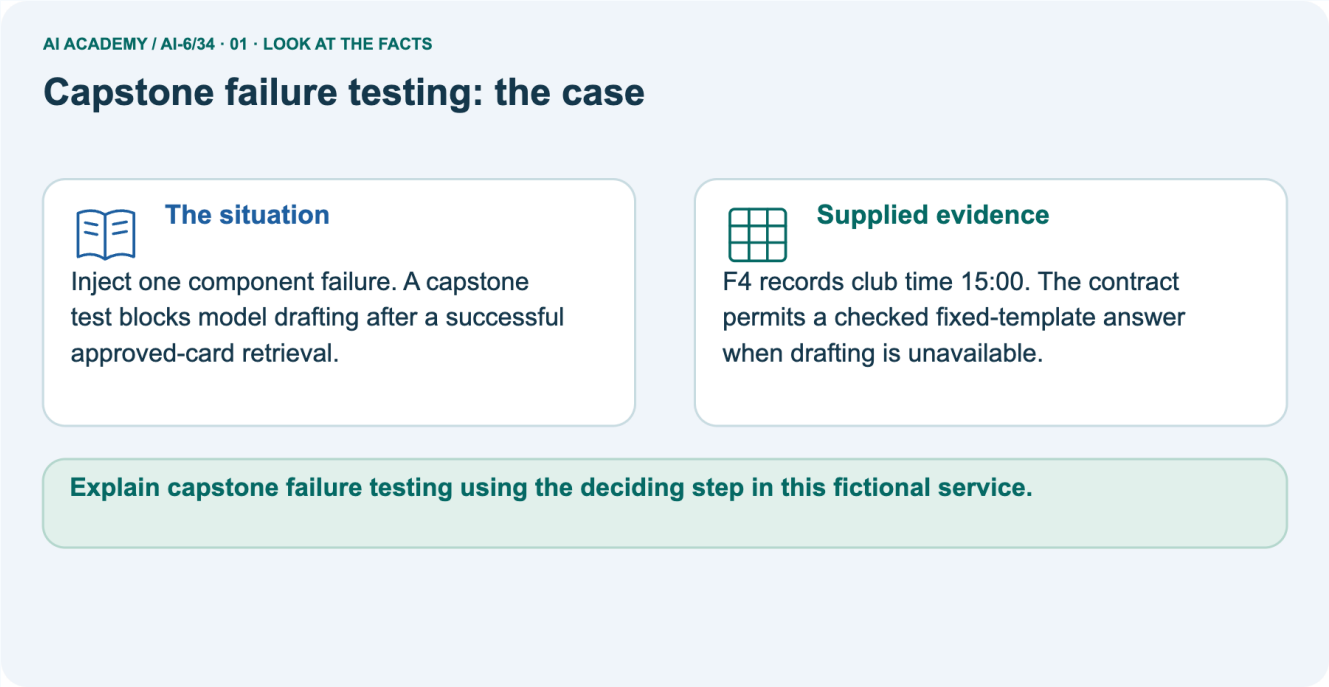
▼ **Reference · Six-picture response sheets**

REFERENCE · SECTION 15

Reference · Six-picture response sheets

Choose the picture that helps explain the current step.

01. Capstone failure testing: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

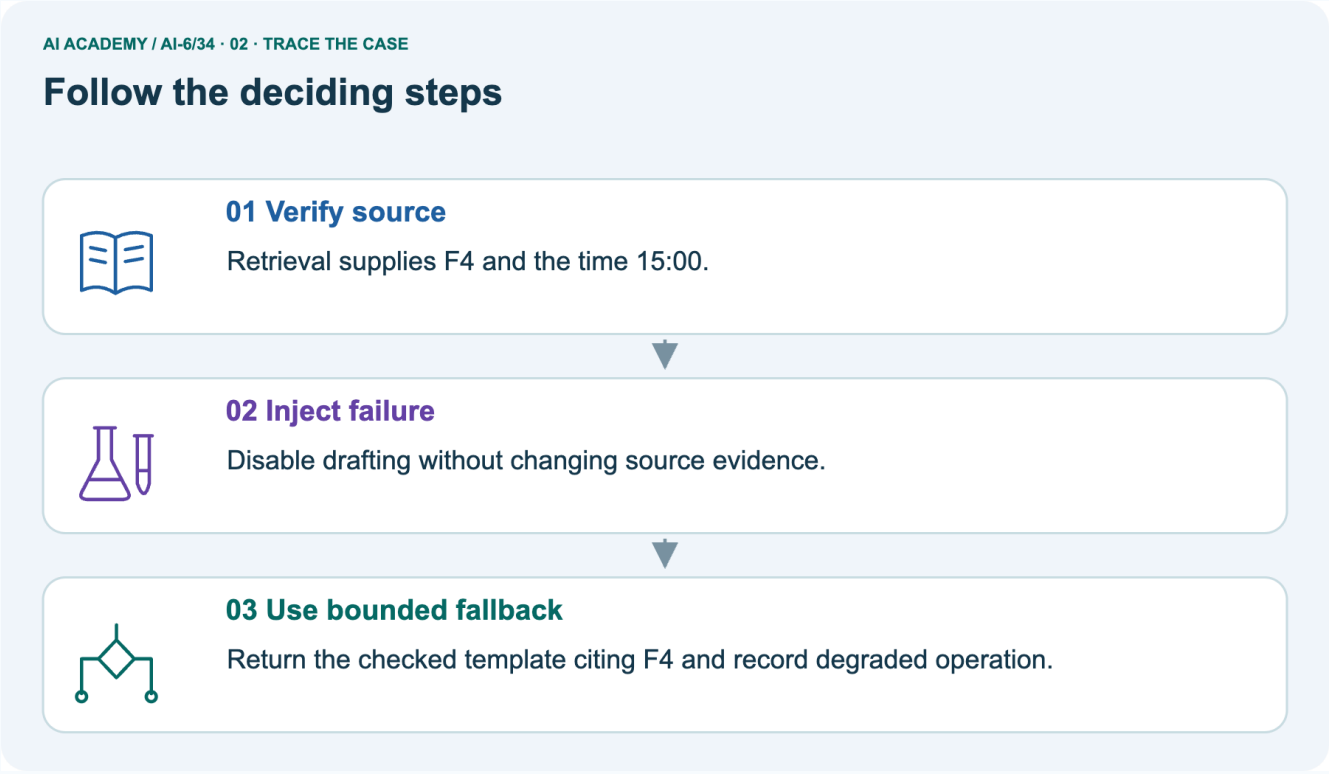
Inject one component failure. A capstone test blocks model drafting after a successful approved-card retrieval. Supplied facts: F4 records club time 15:00.

The contract permits a checked fixed-template answer when drafting is unavailable.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



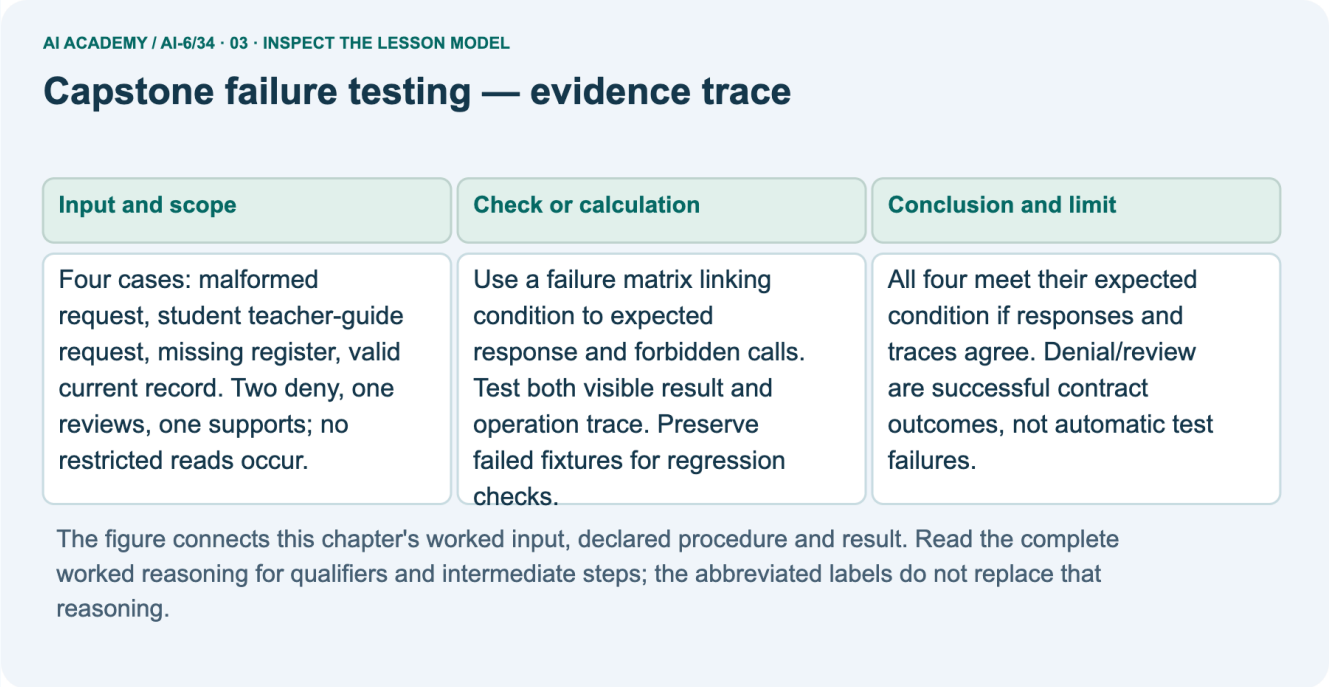
Follow the labelled picture with the matching explanation. Open the vector diagram

Verify source: Retrieval supplies F4 and the time 15:00. → Inject failure: Disable drafting without changing source **evidence**. → Use bounded fallback: Return the checked template citing F4 and record degraded operation.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Capstone failure testing — evidence trace



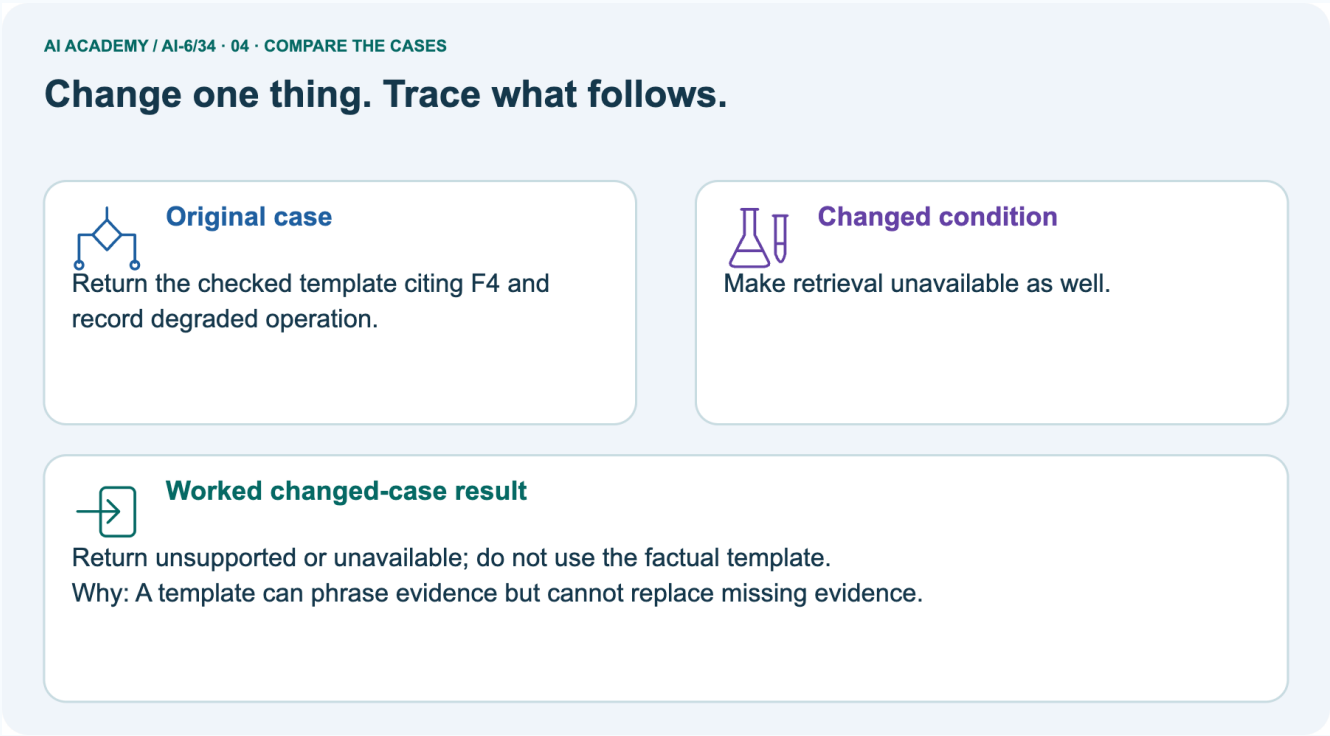
Follow the labelled picture with the matching explanation. Open the vector diagram

Capstone failure testing — **evidence** trace. Four cases: malformed request, student teacher-guide request, missing register, valid current record. One rejects invalid input, one denies access, one reviews and one supports; no restricted reads occur. | Use a **failure matrix** linking condition to expected response and forbidden calls. Test both visible result and operation trace. Preserve failed fixtures for regression checks. | All four meet their expected condition if responses and traces agree. Denial/review are successful contract outcomes, not automatic test failures.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

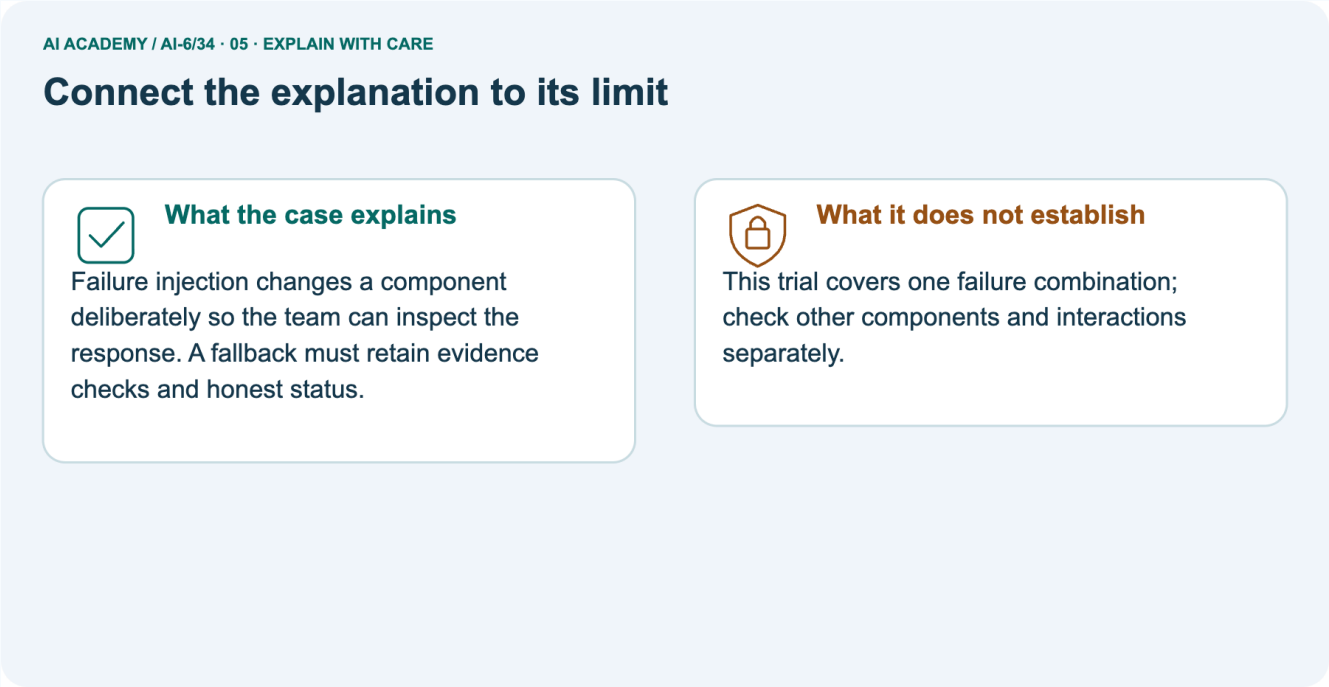
Original case: Return the checked template citing F4 and record degraded operation. Changed condition: Make retrieval unavailable as well. Expected result: Return unsupported or unavailable; do not use the factual template.

Reason: A template can phrase **evidence** but cannot replace missing **evidence**.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

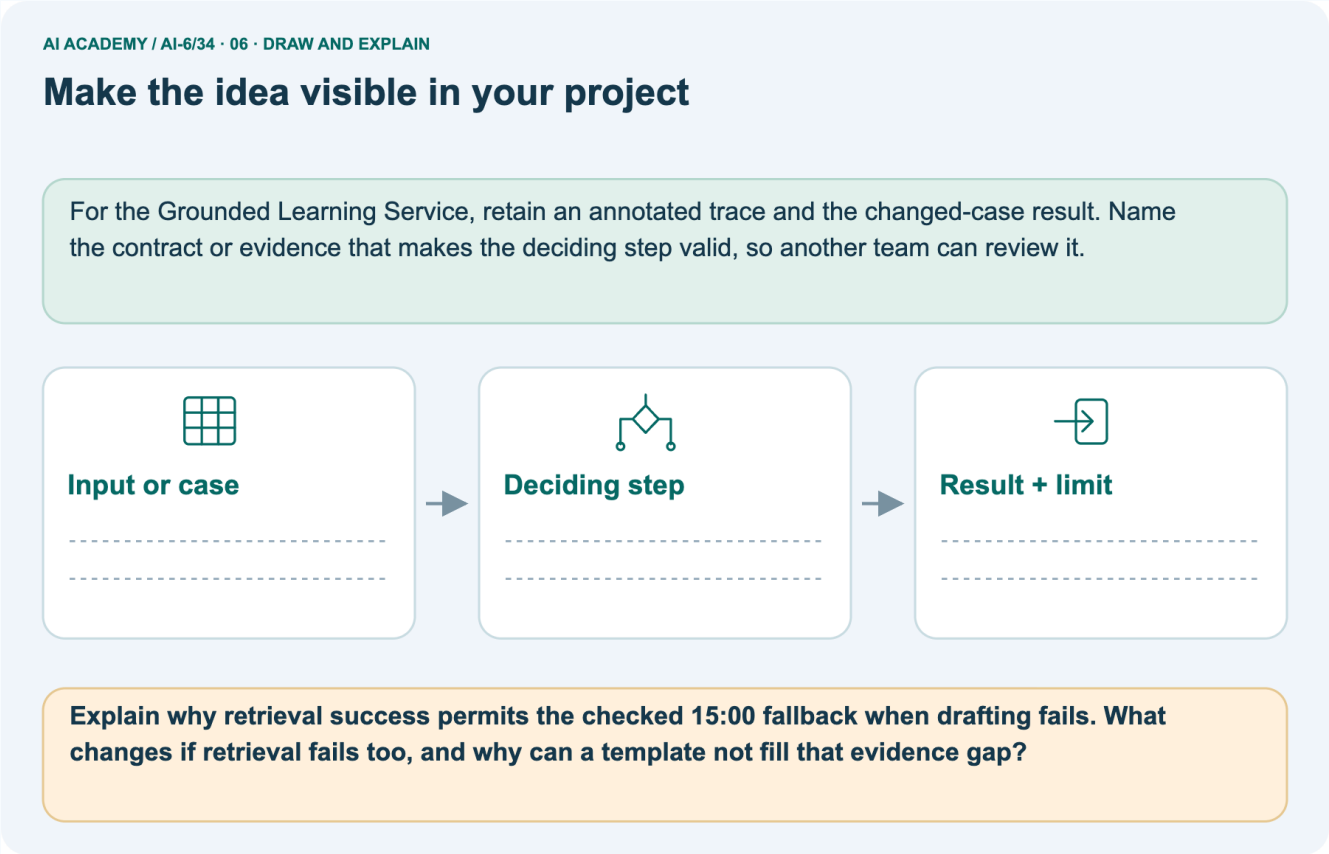
Failure injection changes a component deliberately so the team can inspect the response. A fallback must retain **evidence** checks and honest status.

Limit: This trial covers one failure combination; check other components and interactions separately.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Draw your **input**, deciding step, result and limit. Supported question: Explain why retrieval success permits the checked 15:00 fallback when drafting fails.

What changes if retrieval fails too, and why can a template not fill that **evidence** gap?

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Capstone failure testing: the case

02. Follow the deciding steps

03. Capstone failure testing — evidence trace

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.